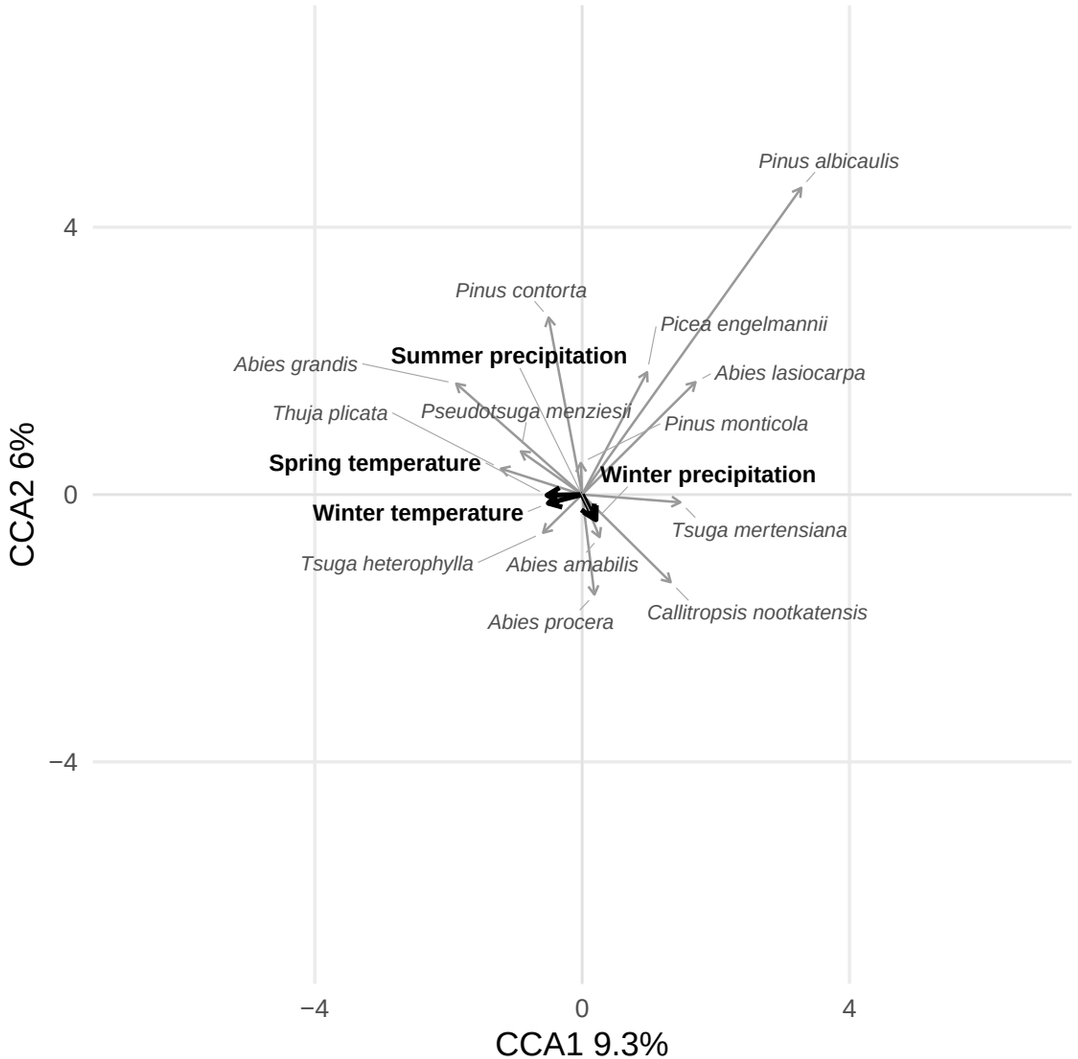
The same CCA under the two scalings

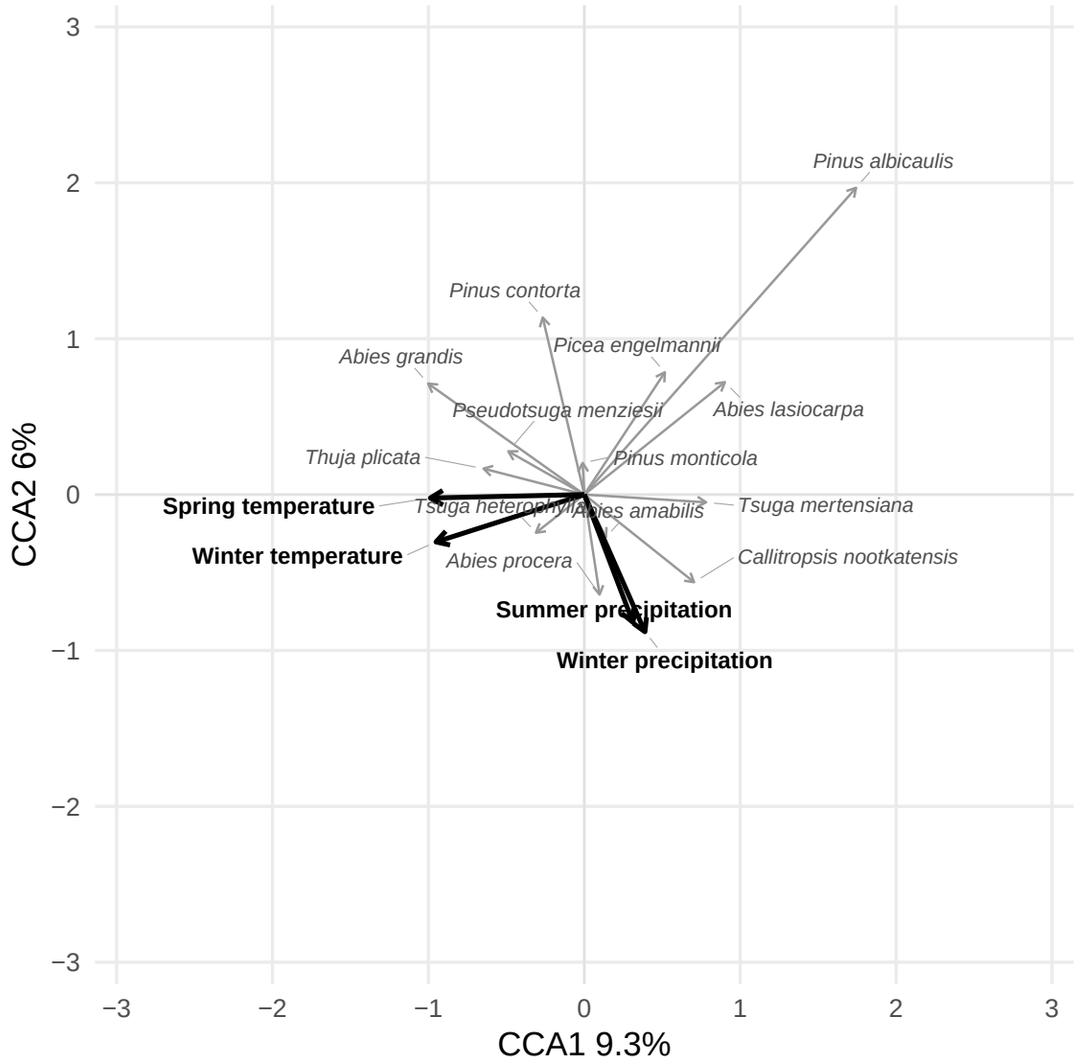
scaling = "sites" (vegan scaling 1)

site scores x eigenvalues; species unscaled
a SAMPLE sits at the weighted centroid of its SPECIES



scaling = "species" (vegan scaling 2)

species scores x eigenvalues; sites unscaled
a SPECIES sits at the centroid of the SAMPLES holding it



Black = climate constraints, grey = species. Raw scores, no display multiplier, so relative lengths are as computed. Axes are drawn to each panel's own range; note the x/y tick values differ. Species direction, order and sign along each axis are IDENTICAL in the two panels − only lengths change (species are x0.532 on CCA1 and x0.429 on CCA2 in the right panel relative to the left). Under scaling 1 the climate arrows shrink by the same factors instead.